Algorithms Notes

- - Heaps
- binary heap - called a heap - its a binary tree
- max heap - largest value at top
- stored as array, usually starts at element 1
- Parent(4) = 2 (indexes)
- Parent(i) = i/2 (with integer division, decimal is dropped)
- left child of i = 2i
- right child of i = 2i+1
- key is what is stored
- min heap has minimum value on top
- to insert, add at the end
- need the array + the size of the heap
- Extract-Max - deletes top node, tree is then restructured,
  used to convert a heap to a sorted list
- - Binomial Heaps
- a collection of binomial trees
- B_k has 2^k nodes
- levels represent combinations
- convert # of nodes to binary to see which trees to make
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